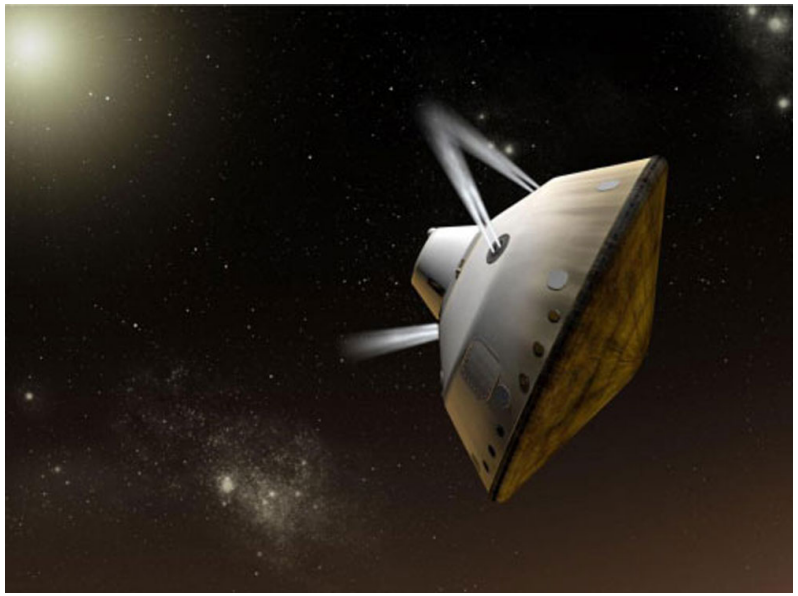
Sparse Identification for Stiff ODE with Chemical Reactions through Physics Informed Neural Network

Xiaoxi Zhou (Lexie)



Motivation

$10^{-3}s$ $10^{-9}s$

Continuity

$$\frac{\partial \rho_s}{\partial t} + \nabla \cdot (\rho_s \boldsymbol{u}) = \nabla \cdot (\rho D_s \nabla Y_s) + \dot{\omega}_s$$

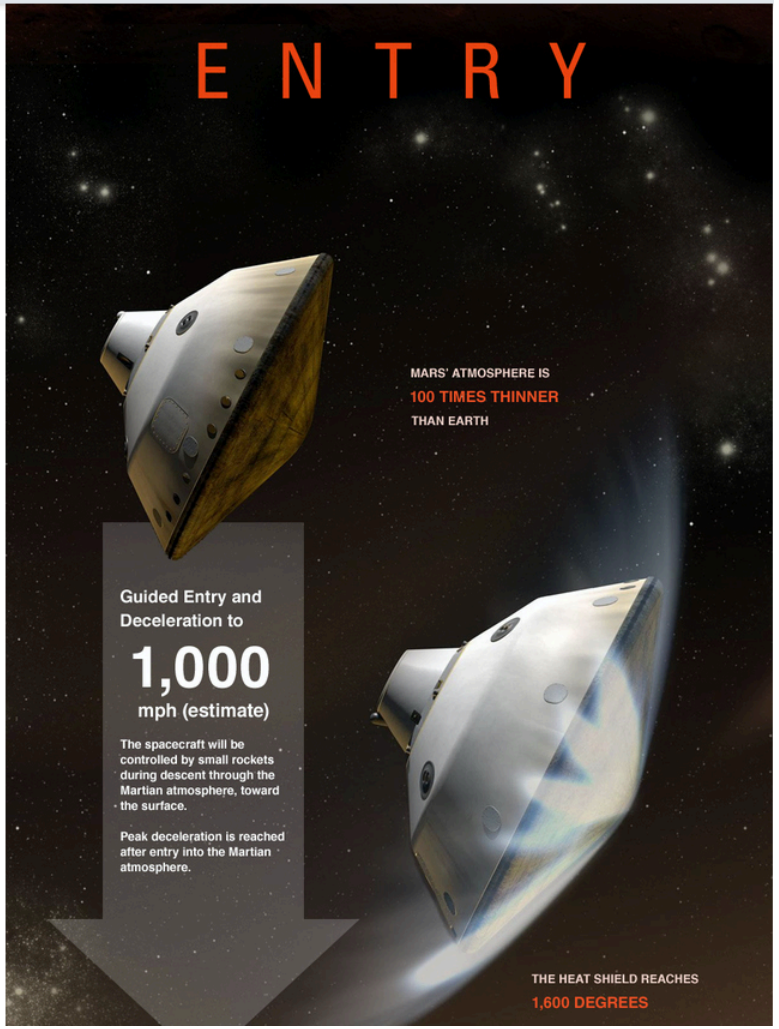
Momentum

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \times \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau}$$

Energy

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\rho E \mathbf{u}) = \nabla \cdot (k \nabla T) - \sum_s h_s \dot{\omega}_s + \ldots$$

Jet Propulsion Laboratory
California Institute of Technology



Main ODE

$$\frac{dc_s}{dt} = S \cdot r(c)$$

$$\dot{\omega}_s = M_s \cdot \left(\frac{dc_s}{dt}\right)$$

$\dot{\omega}_s$ - formation rate of component s $(kg/m^3/s)$

c_s - concentration of component s (mol/m^3)

M_s - mol mass of component s (kg/mol)

For example:

$$CO_2 + O \rightleftharpoons O_2 + CO$$

$$S = [CO_2, O, O_2, CO]^T = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\frac{d[CO_2]}{dt} = -r_1, \quad \frac{d[O]}{dt} = -r_1, \quad \frac{d[O_2]}{dt} = +r_1, \quad \frac{d[CO]}{dt} = +r_1$$

$$r_1 = k_f[CO_2][O] - k_b[O_2][CO]$$

Arrhenius law: $k_f = A \cdot T^n \cdot \exp\left(-\frac{E_a}{RT}\right)$

Related work

	Paper	Advantage	Disadvantage
Ji, Qiu, Shi, Pan, Deng (2020)	Stiff-PINN	Making PINN trainable through Quasi-Steady-State Assumption	1. Need to identify which species are suitable for QSSA 2. QSSA made some approximation assumptions
Champion, Lusch, Kutz, Brunton (2019)	SINDy-autoencoder	Representing complex dynamical system with sparse polynomials	Cannot process well ODEs like $\dot{x} = \frac{f(x)}{g(x)}$
Kaheman, K., Kutz, J. N., Brunton (2020)	SINDy-PI	Turn such ODEs back to how SINDy can process it mathematically	Depends on theta function package (no exp() term)
Maas & Pope (1992)	Intrinsic Low-Dimensional Manifolds for chemical reaction function	Chemical system will go to low dim manifold due to many quick reactions	Only consider the case of pure chemical reaction (don't know if fluid P/T will affect)

Related work

Need more related work that can solve the chemical ODE directly

(Maybe try some other method that is not NN)

ML methods

machine learning:

generalized linear regression

(No nonlinear term)

support vector machine

(No time evolution)

random forest

(Cannot differentiate, but can be used to determine which species is important)

deep learning:

MLP: overfitting

CNN: graph

GNN: graph

RNN; GRU; LSTM: too slow

transformer: no physics governing (but still can consider)

PINN: can try with QSSA

DeepONet(neural operator): can try, mainly learn the function

SINDy: find the function

Overview

Input: 8 species (X_CO2, X_O2, X_N2, X_CO, X_NO, X_C, X_O, X_N) + T + P + Δt

Output: 8 species (X_CO2, X_O2, X_N2, X_CO, X_NO, X_C, X_O, X_N) + T + P

MLP: performed well, but overfitting

PINNs: MLP (the most basic neural network) + physics governing equation

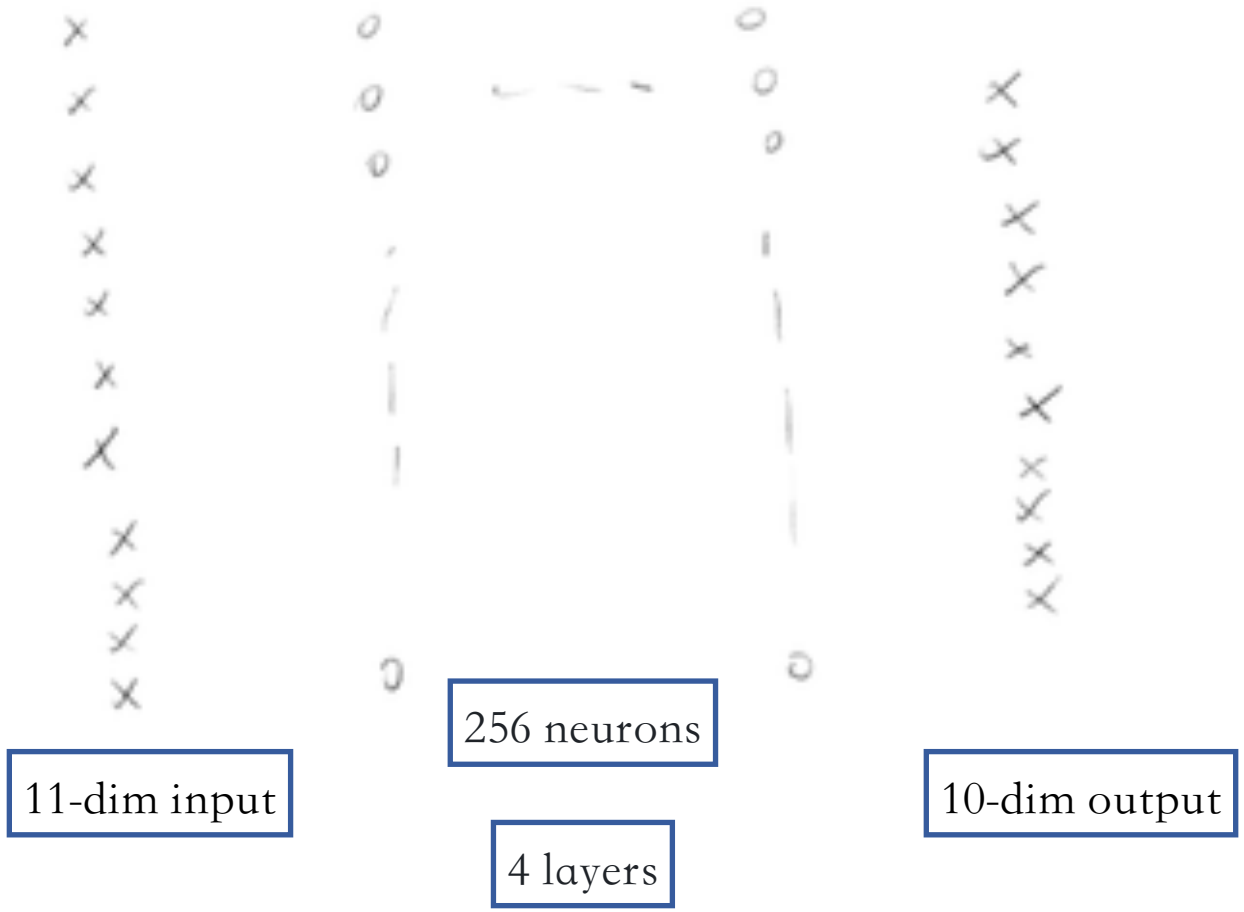
(atom conservation, energy conservation, concentration sums up to 1)

Performed poorly

SINDy + Auto-encoder: sparse identification of nonlinear dynamics; (from my understanding) is to use a sparse polynomial to describe/approximate the whole system the derivative in the ODE of chemical concentration function.

Performed very well, but too slow.

MLP architecture



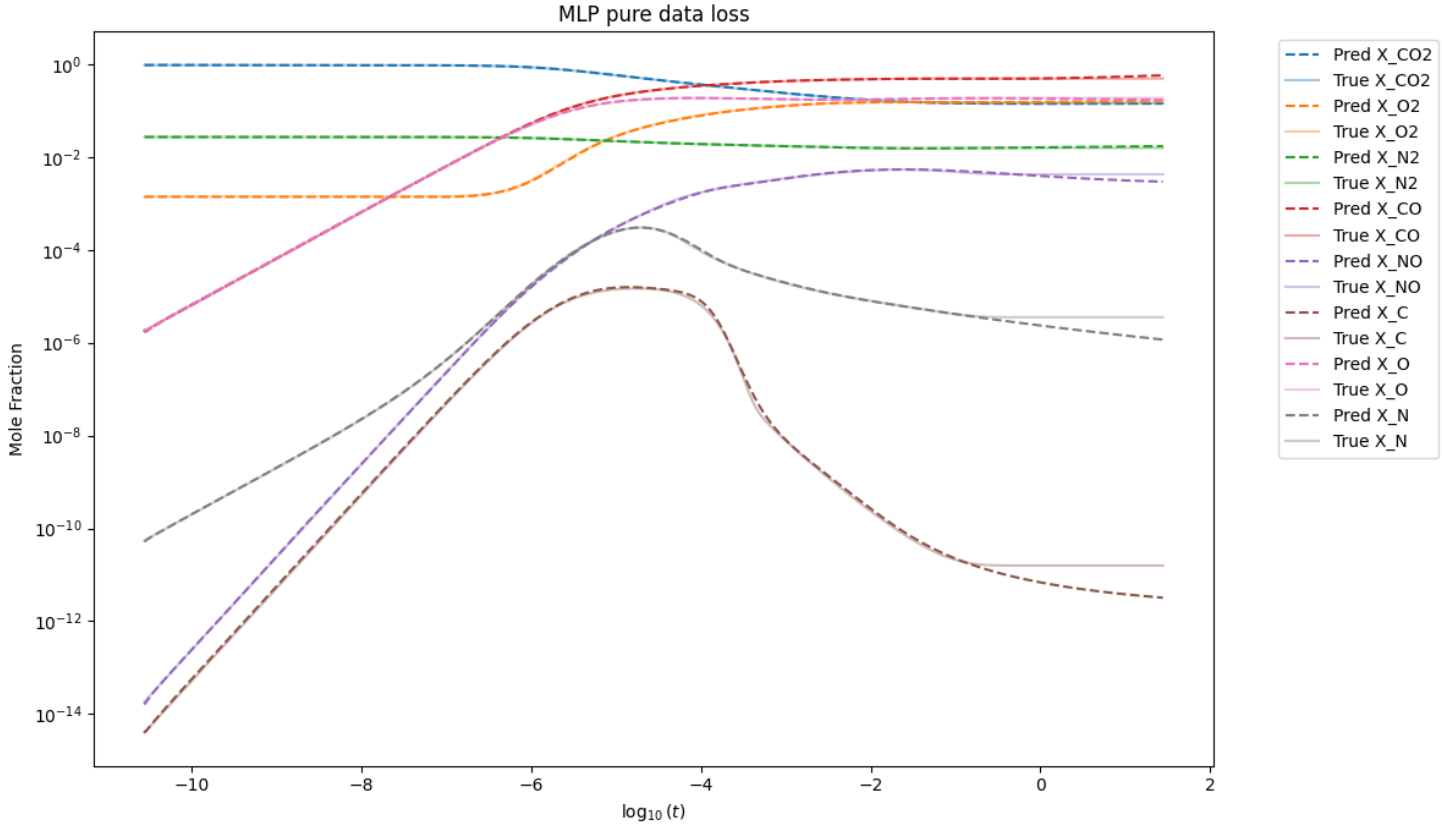
Loss function

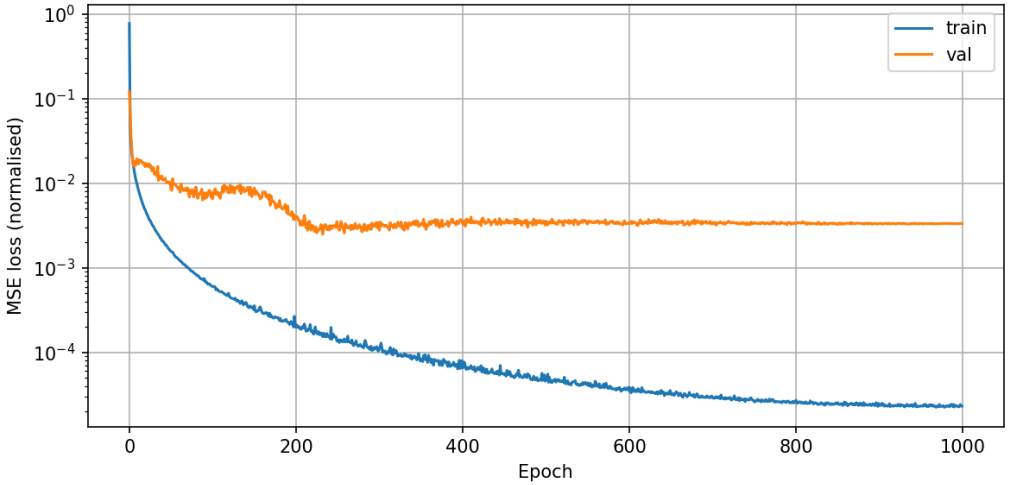
$$\mathscr{L} = \frac{1}{N}\sum_{i=1}^{N}(y_{\text{pred},i} - y_{\text{true},i})^2$$

Optimizer

AdamW

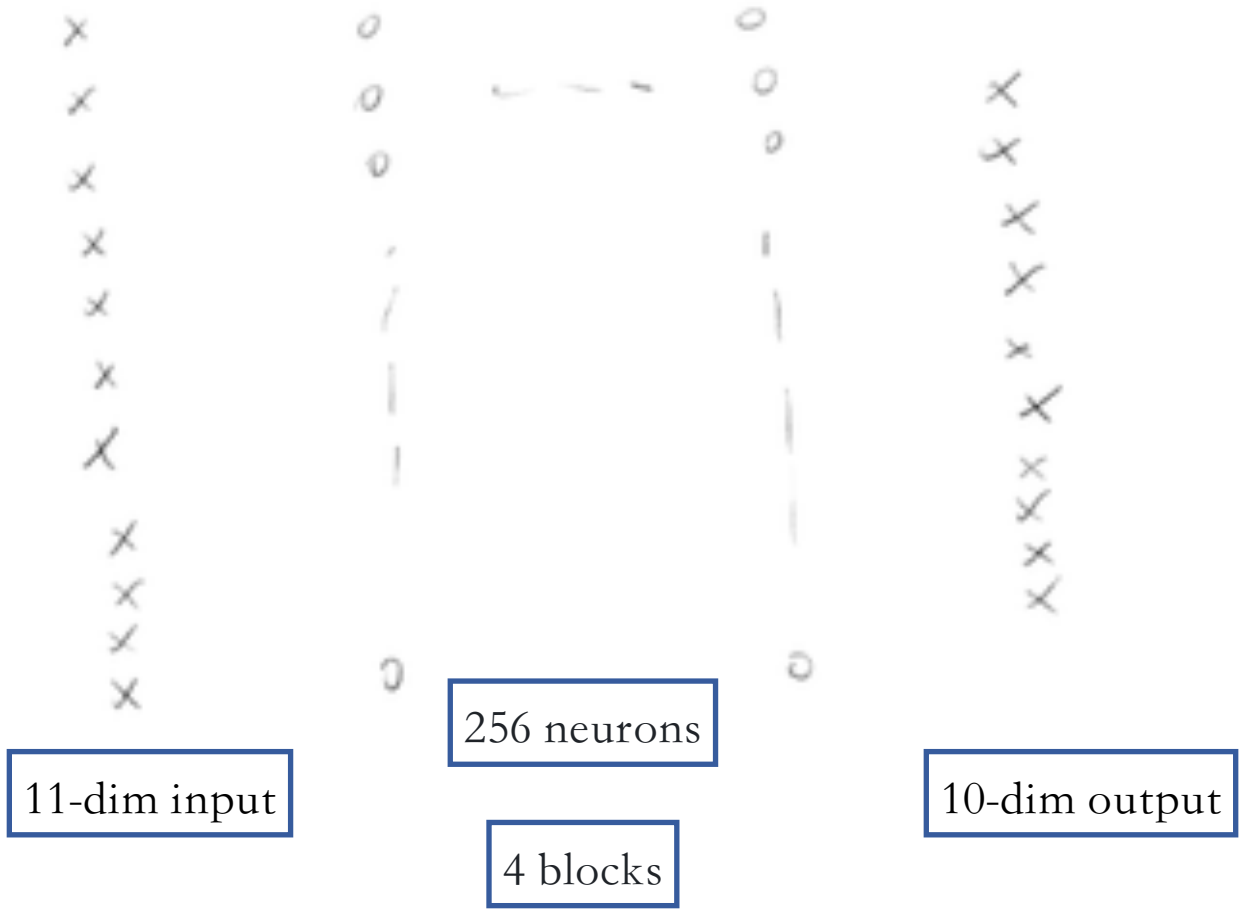
MLP result (pure data loss)





PINN architecture

$$\mathscr{L} = \lambda_{sum} L_{sum} + \lambda_{atom} L_{atom} + \lambda_{energy} L_{energy}$$



Concentration conservation:

$$L_{sum} = (\sum_i X_i - \sum_i X_i^{(0)})^2$$

$$\lambda_{sum} = 1.0$$

Atom conservation:

$$|A\mathbf{X} - A\mathbf{X}^{(0)}|^2$$

$$\lambda_{atom} = 1.0$$

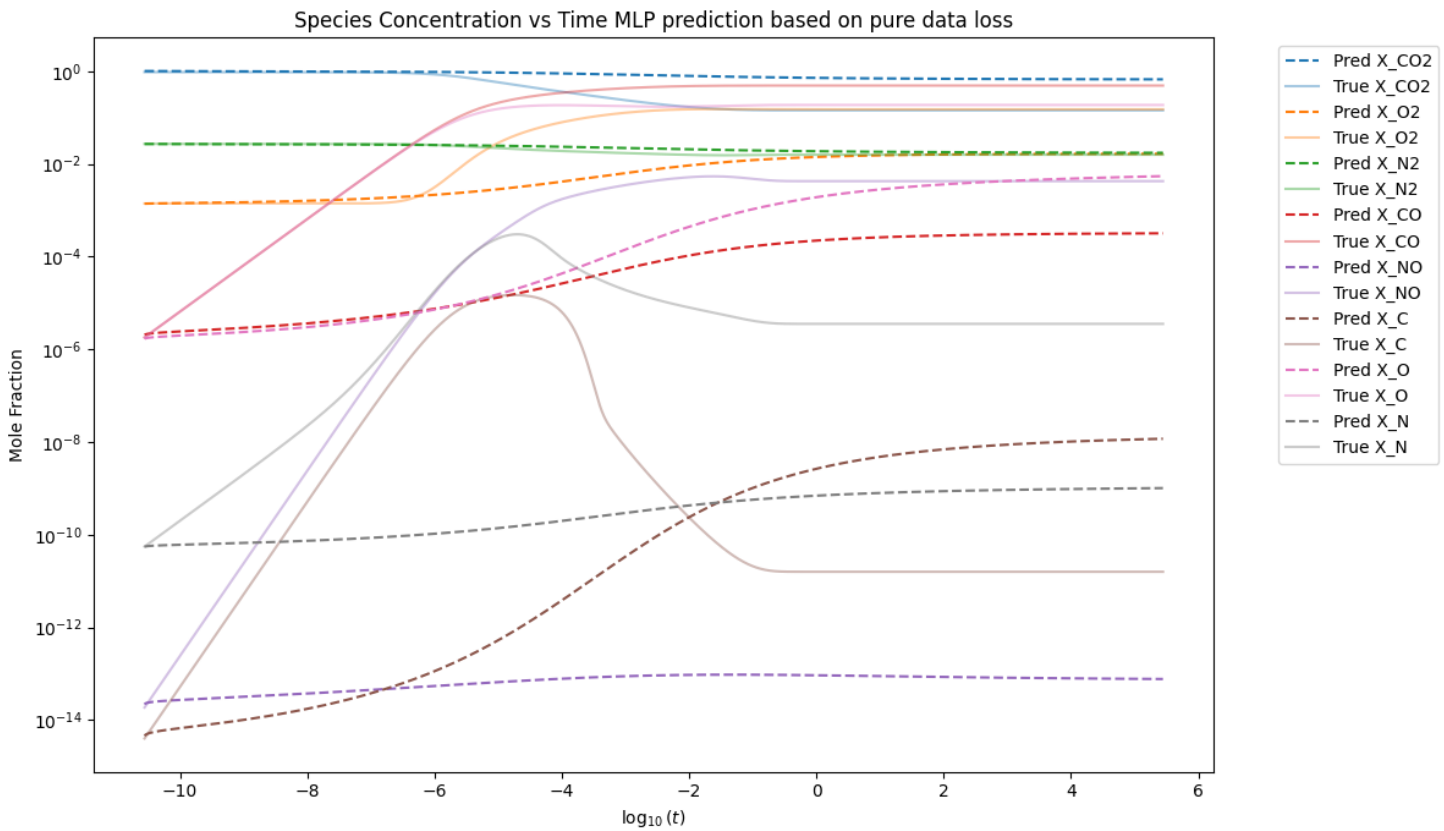
A is 3(C,O,N)*8 atom matrix

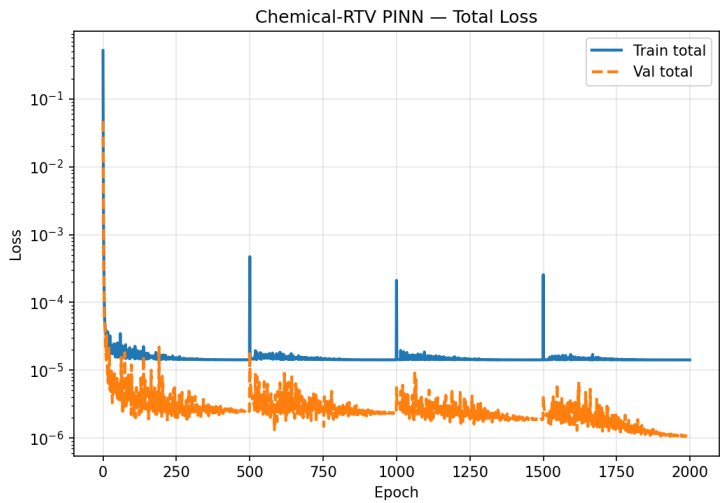
Energy conservation:

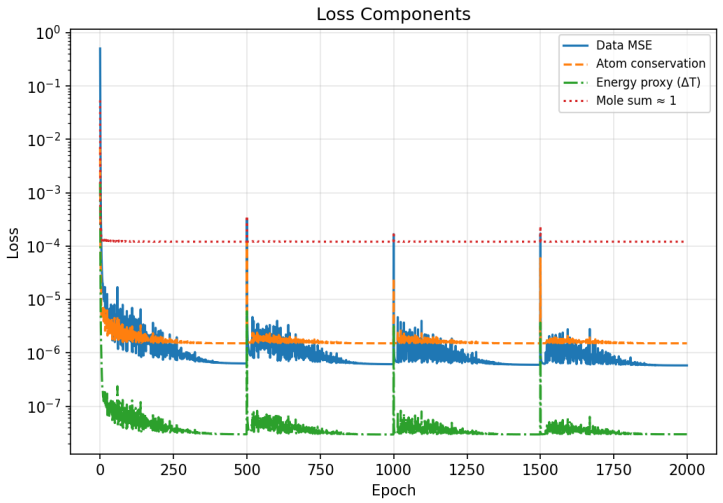
$$\frac{(U_{\text{pred}} - U^{(0)})^2}{U^{(0),2}}$$

$$\lambda_{energy} = 1e - 10$$

PINN result (pure physics loss)

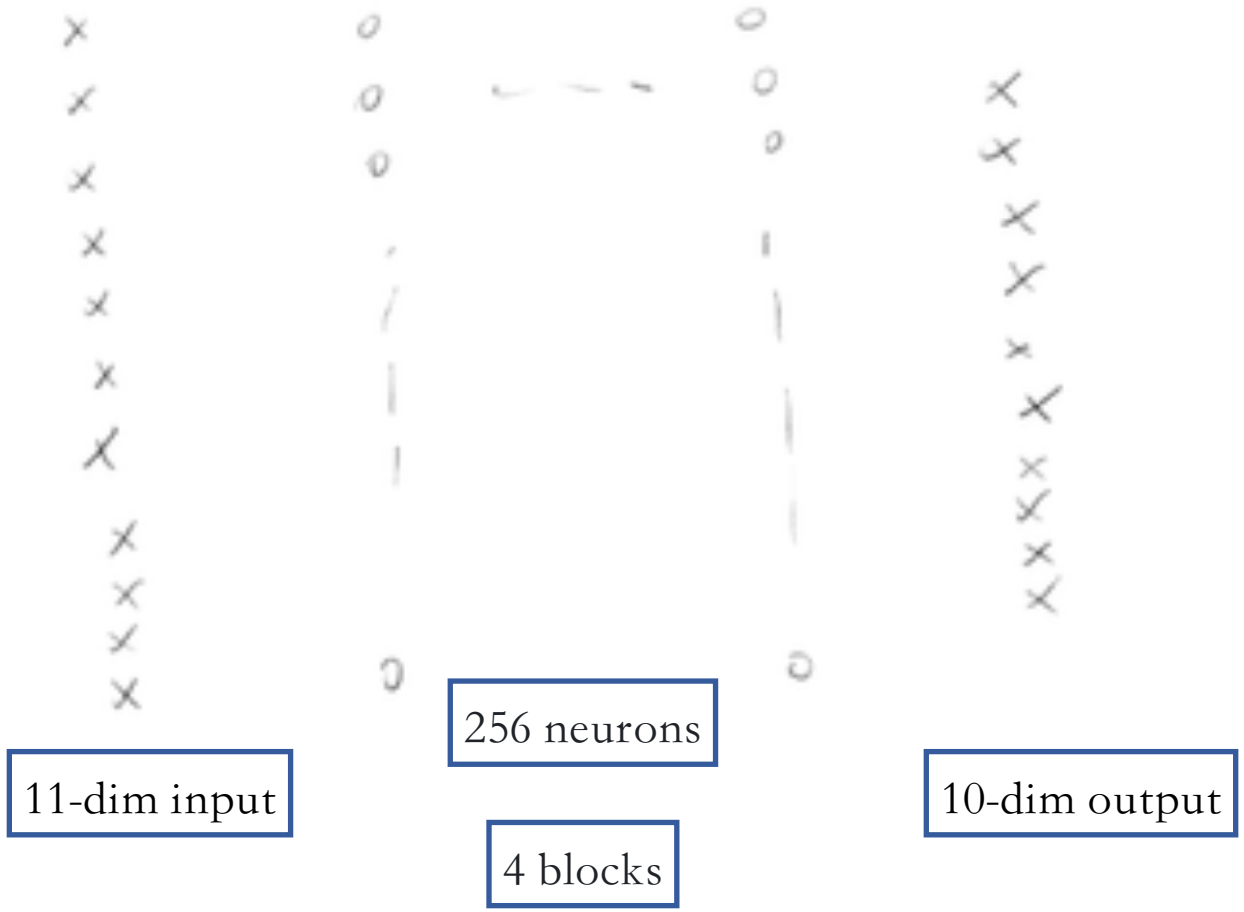






PINN architecture

$$\mathscr{L} = \lambda_{sum} L_{sum} + \lambda_{atom} L_{atom} + \lambda_{energy} L_{energy} + \lambda_{data} L_{data}$$



Concentration conservation:

$$L_{sum} = \left(\sum_i X_i - \sum_i X_i^{(0)}\right)^2, \lambda_{sum} = 1.0$$

Atom conservation:

$$|A\mathbf{X} - A\mathbf{X}^{(0)}|^2, \lambda_{atom} = 1.0$$

A is 3(C,O,N)*8 atom matrix

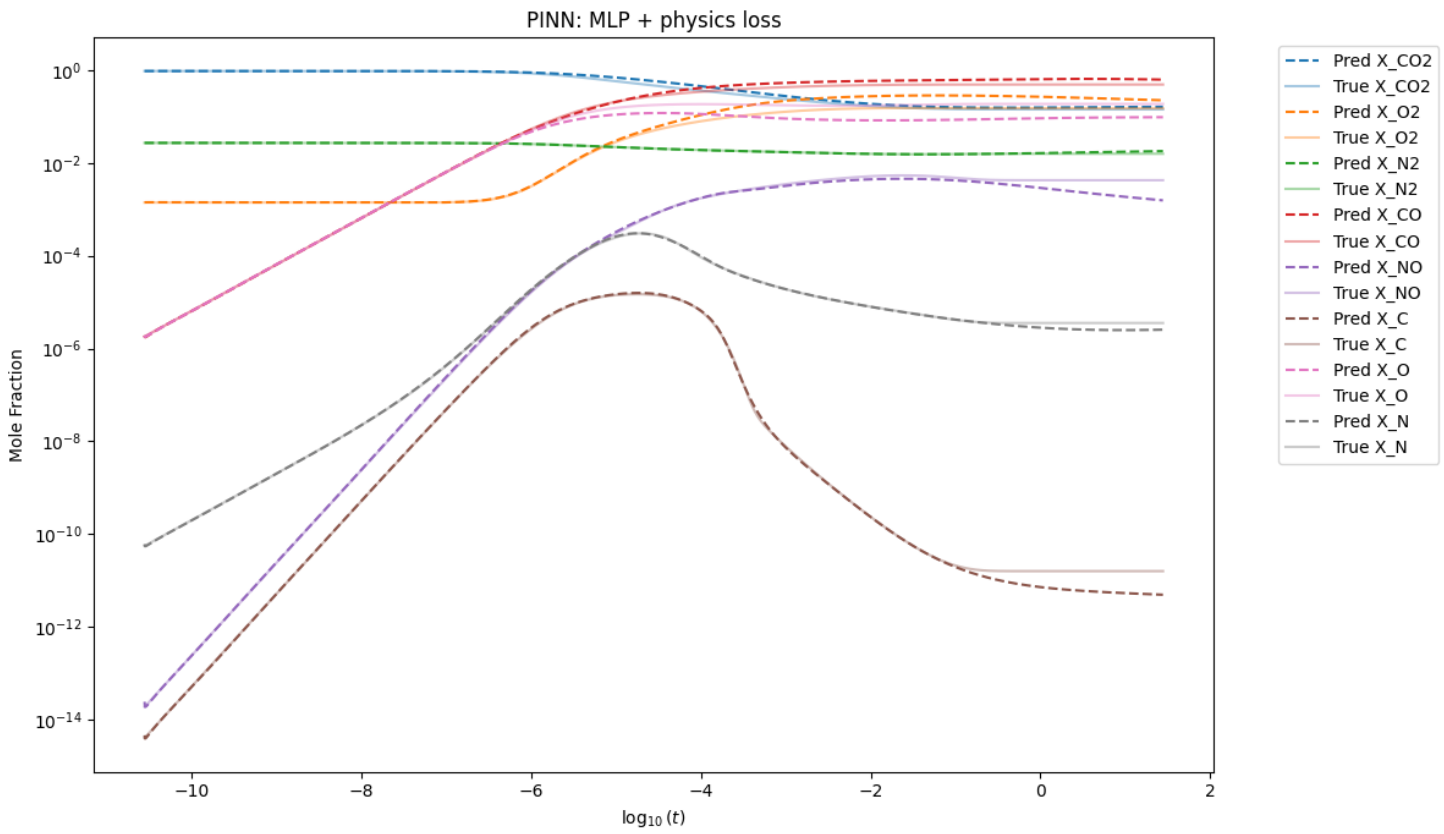
Energy conservation:

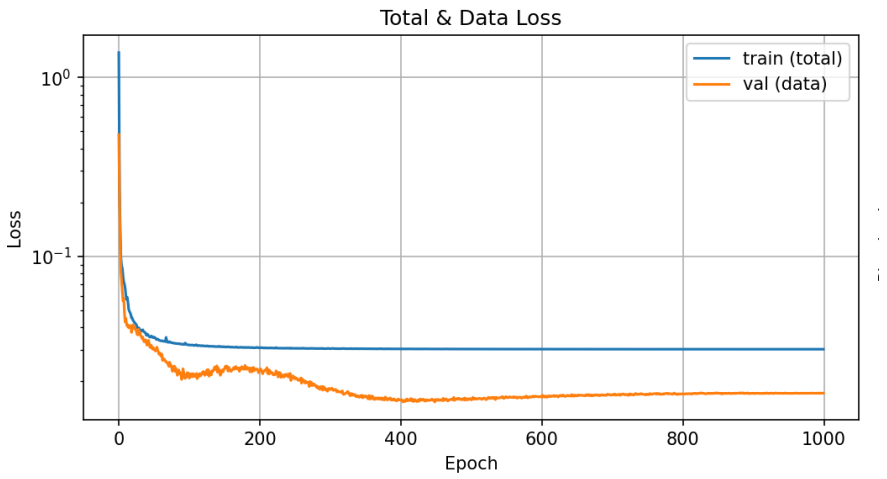
$$\frac{(U_{\text{pred}} - U^{(0)})^2}{U^{(0),2}}, \lambda_{energy} = 1e-10$$

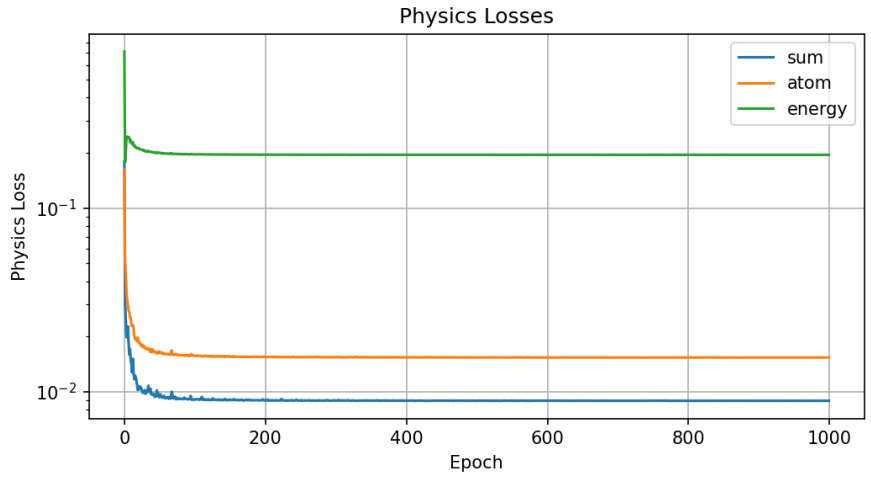
Data loss:

$$\frac{1}{N}\sum_{i=1}^{N} (y_{\text{pred},i} - y_{\text{true},i})^2, \lambda_{data} = 1.0$$

PINN result (with data loss)







Why can we use sparse identification?

Inputs (10/11 variables):

8 species (X_CO2, X_O2, X_N2, X_CO, X_NO, X_C, X_O, X_N) + T + P

Physics governing equation:

Atom conservation: 5 equations (C,O,..)

Energy: 1 equation

Sum of mol = 1: 1 equation

10-7=3

3 degrees of freedom

So theoretically we have this 3-dim manifold that definitely holds our solution

SINDy + auto-encoder architecture

Auto-encoder: 10 dim - 8 dim - 10 dim

φ: [10] → 64 → 64 → 32 → [8] # 3 hidden layers to compress by 2 dimensions

ψ: [8] → 32 → 64 → 64 → [10] # 3 hidden layers to expand by 2 dimensions

Sparse identification loss:

X High dim(10)	φ Encoder	Z Low dim(8)	ψ Decoder	$\hat{x} = \psi(z)$

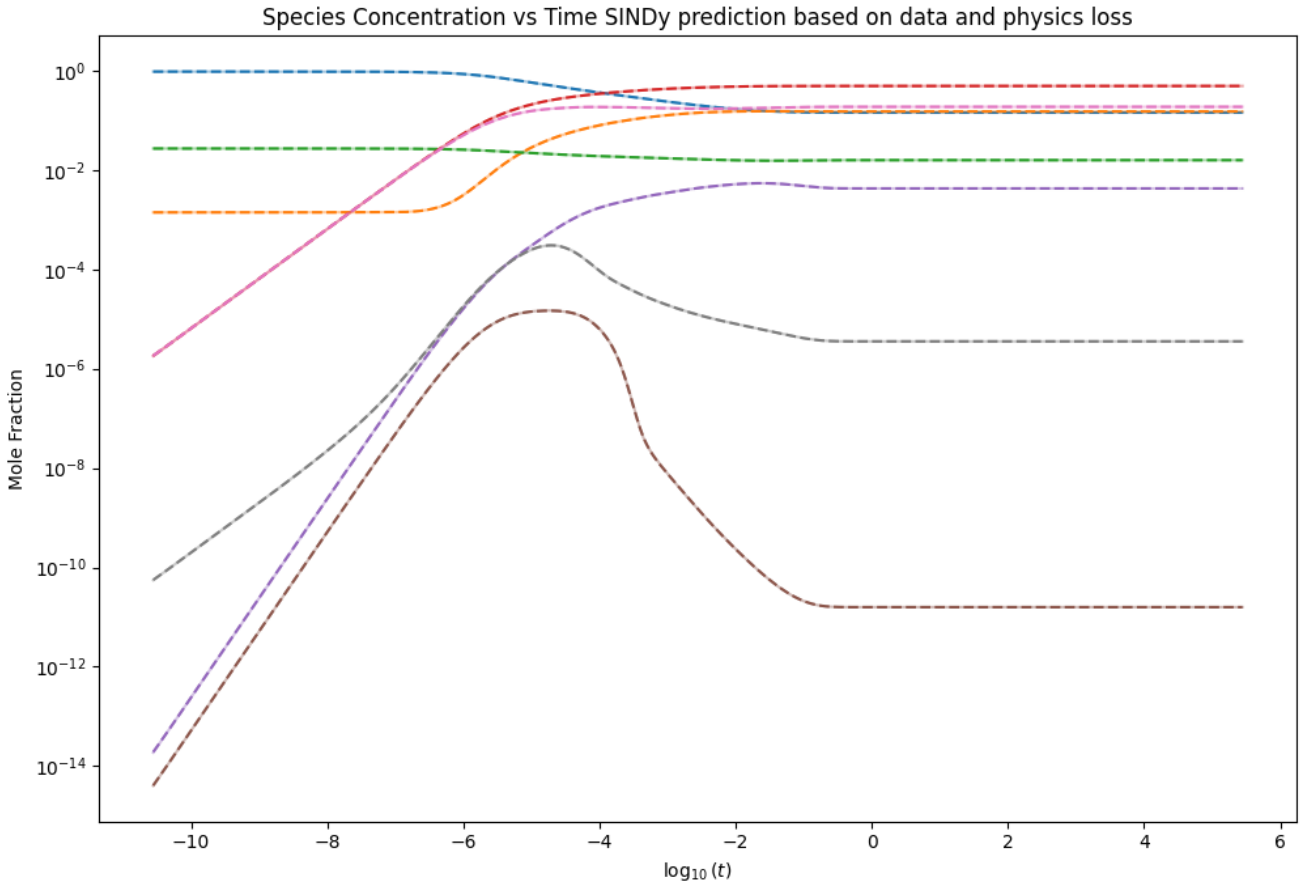
$$\theta(z')\Xi \rightarrow dz/dt$$

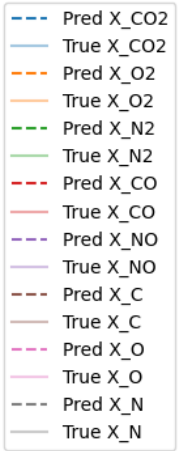
$L_1 = x - \hat{x}$ (For obvious reason lol)

$L_2 = \|\dot{x} - (\nabla_u \psi)(\nabla_u \psi)\Theta(z^T)\Xi\|^2$ Derivative of x and z space should be the same $L_3 = \|(\nabla_x z)\dot{x} - \Theta(z^T)\Xi\|^2$

Physics loss: Concentration add up to 1; Atom number conservation; Energy conservation

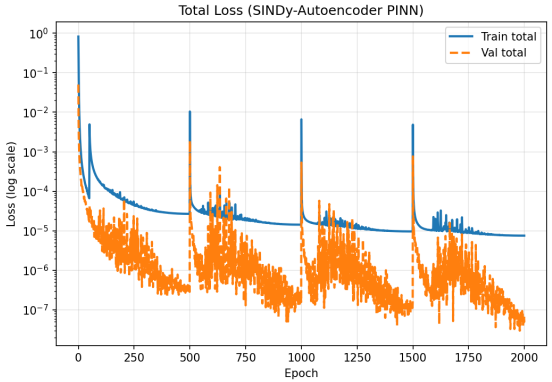
Discussion - SINDy with data and physics loss

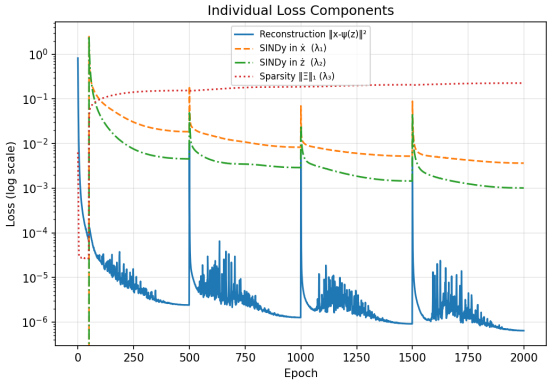




SINDy: solving the chemical reactions with polynomials ☠️

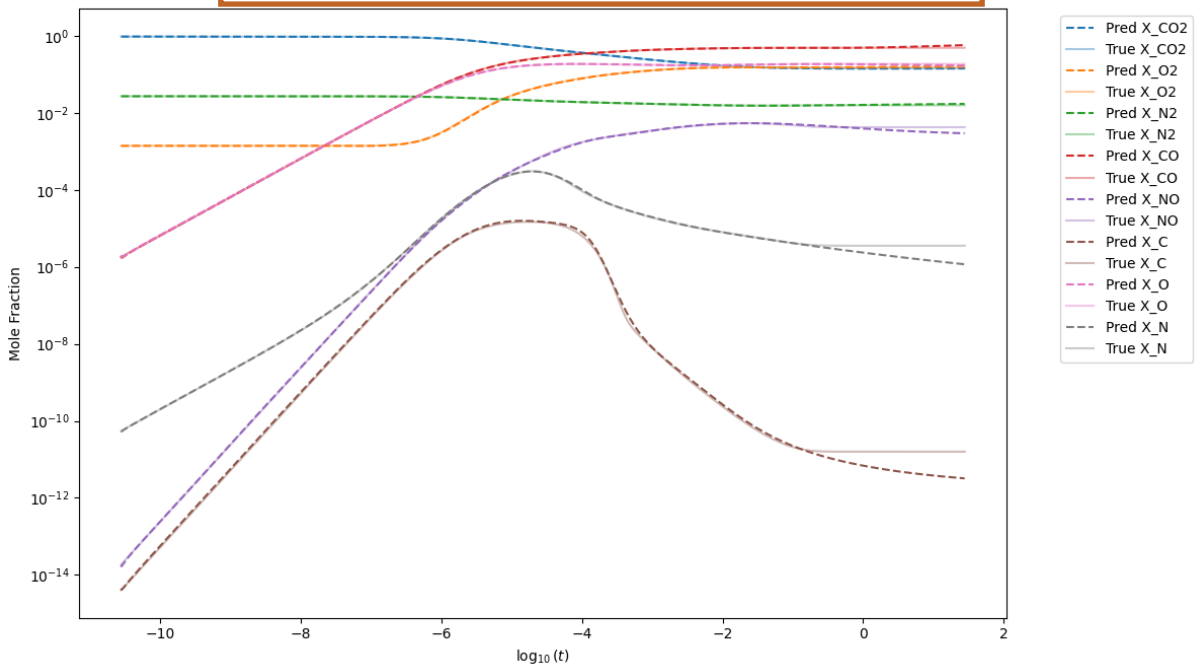
$$
\begin{aligned}
\dot{z}_1 = &+0.2975 \cdot 1 - 0.4070 \cdot z_1 + 0.3378 \cdot z_2 - 0.3849 \cdot z_3 \\
&+0.1414 \cdot z_4 - 0.3665 \cdot z_5 + 0.4068 \cdot z_7 - 0.0971 \cdot z_8 \\
&-0.5193 \cdot z_1^2 - 0.7782 \cdot z_1 z_2 + 0.5447 \cdot z_1 z_3 - 0.0980 \cdot z_1 z_4 \\
&-0.1742 \cdot z_1 z_5 - 0.4028 \cdot z_1 z_6 - 1.3107 \cdot z_1 z_7 + 0.6289 \cdot z_1 z_8 \\
&+0.0619 \cdot z_2^2 - 0.2821 \cdot z_2 z_3 + 0.2093 \cdot z_2 z_4 + 0.1894 \cdot z_2 z_5 \\
&+0.2642 \cdot z_2 z_6 - 0.3329 \cdot z_2 z_8 + 0.2557 \cdot z_3^2 - 0.3262 \cdot z_3 z_4 \\
&+0.4917 \cdot z_3 z_5 - 0.6696 \cdot z_3 z_7 + 0.2470 \cdot z_3 z_8 + 0.0631 \cdot z_4 z_6 \\
&+0.3222 \cdot z_4 z_7 - 0.1614 \cdot z_4 z_8 - 0.8893 \cdot z_5^2 - 0.5511 \cdot z_5 z_6 \\
&+0.3573 \cdot z_5 z_7 + 0.4478 \cdot z_5 z_8 - 0.0951 \cdot z_6^2 + 0.3031 \cdot z_6 z_7 \\
&+0.1249 \cdot z_6 z_8 + 0.6070 \cdot z_7^2 - 0.4831 \cdot z_7 z_8
\end{aligned}
$$



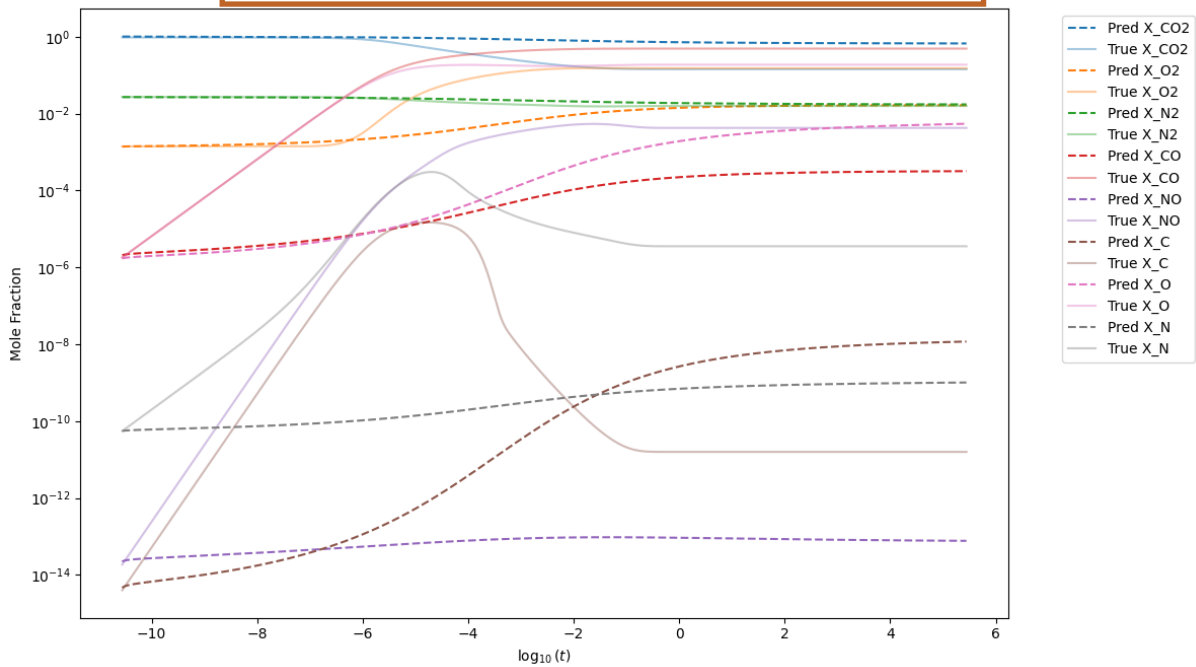


Overall performance

MLP (pure data loss)

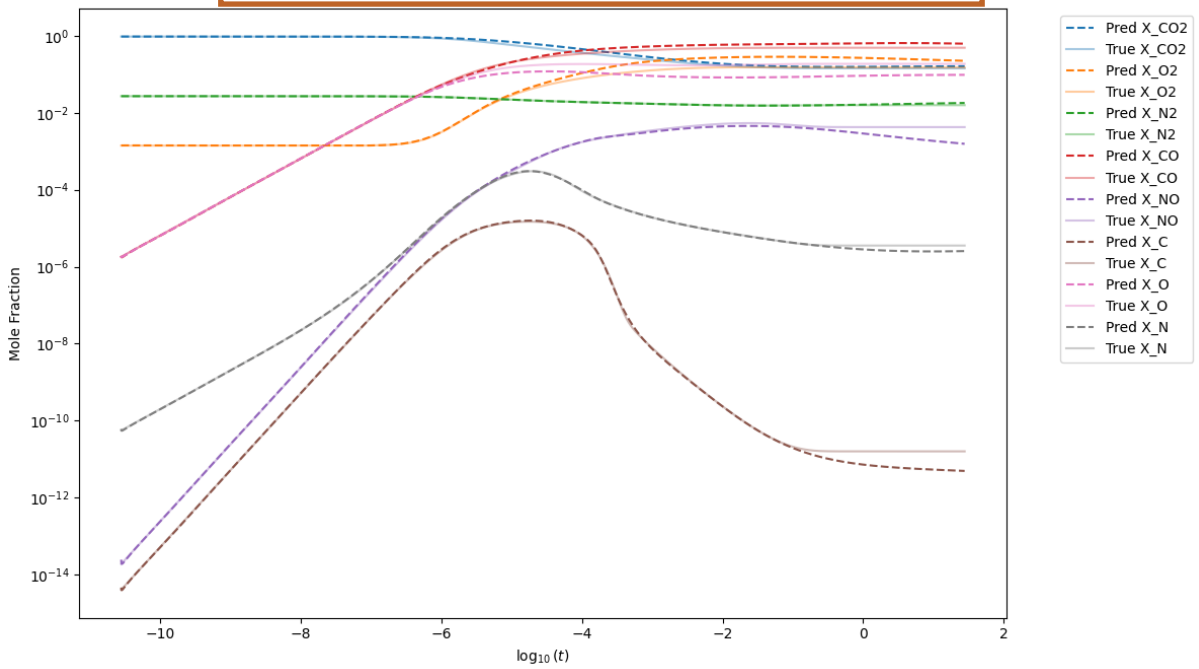


PINN (pure physics loss)

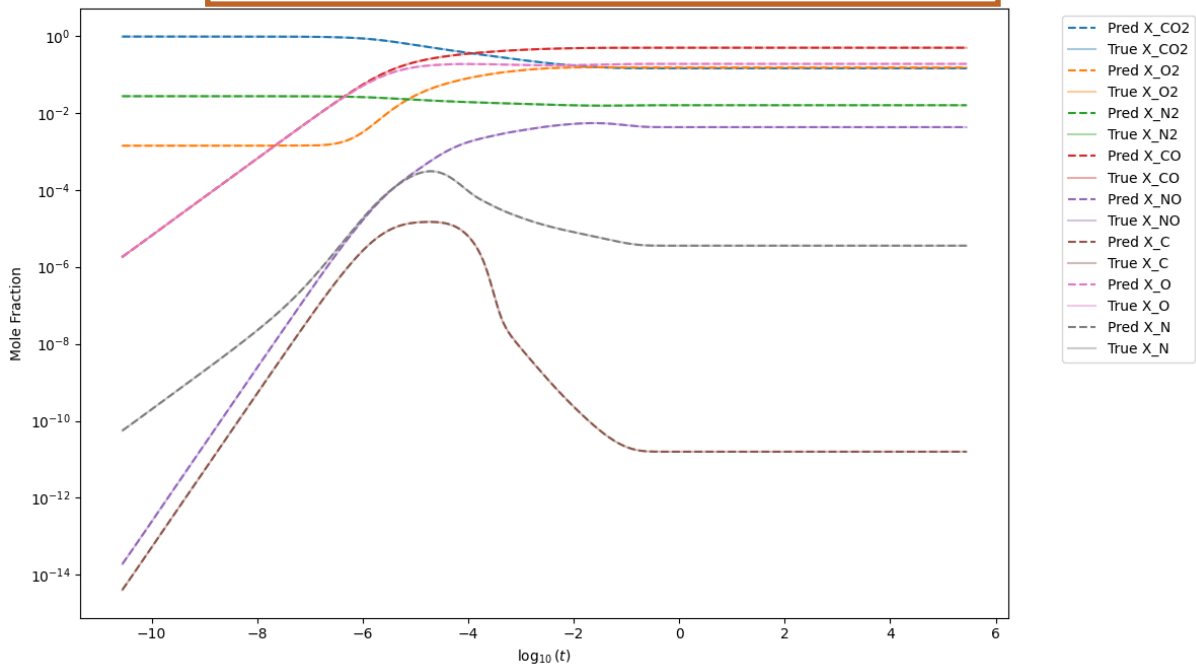


Overall performance

PINN(data + physics loss)



SINDy + auto-encoder



Why can we use sparse identification?

Maas & Pope(1992)

It has been proved that there exists an intrinsic low-dimensional manifold (ILDM) in the component space of complex chemical reactions. The system trajectory will rapidly converge to this low-dimensional surface.

Witney Embedding theorem

Any smooth d-dimensional manifold can be smoothly embedded into a 2d-dimensional Euclidean space.

Universal Approximation theorem

A sufficiently wide single-hidden-layer network (or a sufficiently deep network) can approximate any continuous function on any compact set with arbitrary precision.

Conclusion and future steps

All the methods here iterate through $10^{-9}s$ (so 10^6 times)
The goal here is to acquire directly the result after $10^{-3}s$, without iterating through 10^6 times, with same loss function, same neural network, but predict almost accurate with different initial condition (T, P, initial condition)

Future work:
Generate data with different initial condition.
See if I can write the polynomial with fewer parameters.
(If it does then SINDy method works)
Try out more methods in related work or try different ML methods.

Sparse Identification for Stiff ODE with Chemical Reactions through Physics Informed Neural Network

Xiaoxi Zhou (Lexie)

Thanks for listening!

1 Introduction
Motivation, problem setup

2 Related work
Relative work, why SINDy & PINN

3 Architecture
How does it work and why it works

4 Optimization
Loss functions, why it matters

5 Results
Results, summary and discussion